

Han Yuning

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EDUCATION

UNIVERSITY OF FLORIDA

Ph.D in Computer Science

Gainesville, FL

Aug 2025 - Present

COLUMBIA UNIVERSITY

M.S. in Electrical Engineering

Tesla Scholarship of Columbia University Electrical Engineering Department(Honored)

New York City, NY

Sept 2023 - Feb 2025

SHANGHAI JIAOTONG UNIVERSITY (SJTU)

BS in Information Engineering

The Eastern China's second prize of the Freescale Cup National College Student Smart Car Competition 2020(Top 20%)

Shanghai, CN

Sept 2019 - Jun 2023

RESEARCH EXPERIENCE

GRADUATE RESEARCH ASSISTANT

INFERENCE SPPELUP STRUCTURE DESIGN FOR MEMORY-LIMITED LLM

Gainesville, FL

Advisor: Dr. Jingwei Sun, University Of Florida

Sept 2025 – May 2026

- First author, posted paper: [CATS: Cascaded Adaptive Tree Speculation for Memory-Limited LLM Inference Acceleration](#)
- Designed a self-speculative decoding structure called CATS, applicable for different auto-regressive structure LLMs, and outperformed in SOTA self-speculative decoding structures in inference speed under edge scenarios.
- We evaluate CATS on different models across five benchmarks on real edge devices. CATS can achieve a wall-clock speedup of up to 5.08x with no degradation in generation quality, outperforming the SOTA method by up to 1.45x under edge memory constraints. NeurIPS 2026 under review.

ANALYSIS OF BACKDOOR ATTACK AND PROTECTION OF DIFFUSION MODEL

Boston, MA

Advisor: Prof. Yingjie Lao, Tufts University

April 2024 - Feb 2025

- First author, posted paper: [UIBDiffusion: Universal Imperceptible Backdoor Attack for Diffusion Models](#) in CVPR 2025.
- Applied specific imperceptible trigger to part of images in the dataset to poison it. This method causes the diffusion model, trained on the poisoned dataset, to exhibit significant performance deviations when generating target images upon detecting the trigger, while keeping the quality of image generation in a high level on clean dataset.
- Further explored other trigger construction techniques to perform backdoor attacks on the diffusion model and validated the feasibility of these methods. More experiments will be carried out on TrojDiff model and VillanDiff model.
- Experiments of model training and analysis were carried out on CIFAR10 and CelebA-HQ 256 dataset. Evaluate model performance based on FID score, MSE and SSIM score metrics.

RESEARCH IN REINFORCEMENT LEARNING MODEL BASED ON NON-IDD DATASET

New York City, NY

Advisor: Prof. Javad Ghaderi, Columbia University

Feb 2024 - May 2024

- Using PyTorch to replicate and evaluate the performance of the FAVOR model, which is non-open-source in original paper.
- Implicate the main reinforcement learning part of FAVOR model, which is based on deep Q-learning model, adapt the main RL network into Double DQ network, Dueling DQ network, and designed a new network according to policy gradient method.
- Experiments were carried out on CIFAR-10 dataset, which was generated into four new non-idd datasets based on different non-idd parameter. Gained accuracy close to 49.98% and model reward higher than -0.51, as proposed in the original paper. Made horizontal comparison between original method and our changed RL networks.
- Found that Double DQN method shows best performance in accuracy, reward and model loss, effectively reduces the likelihood of model overfitting and enhances model stability. Dueling DQN showed less stability, according to simple decision network design. Gradient policy method didn't show its superiority in this task.

REPLICATION AND PERFORMANCE EVALUATION OF SWIN TRANSFORMER MODEL

New York City, NY

Advisor: Dr. Mehmet Kerem Turkcan, Columbia University

Oct 2023 - Dec 2023

- Using Tensor Flow to replicate and evaluate the performance of the swin transformer model.
- Implicate the main structure of the swin transformer including core parts listed below: multi-scale processing to extract features, sliding window mechanism, self-attention and multi-head attention mechanisms, masks, and relative positional encoding.
- Experiments were carried out on the tiny image-net dataset. The completed replication of the swin transformer achieved a final experimental accuracy of 99.64%. In contrast, models lacking the shifting window feature dropped to an accuracy of 63.4%, mirroring the characteristic performance decline when key features are missing, as proposed in the original paper.

UNDERGRADUATE RESEARCH ASSISTANT

ADAPTIVE SEMANTIC COMMUNICATION SYSTEM FOR VIDEOS ON MACHINE LEARNING

Shanghai, CN

Advisor: Prof. Zhiyong Chen, Institute Of Wireless Communications, SJTU

Jul 2022 - Sep 2022

- Proposed a hierarchical adaptive semantic communication ML system, applying it to the processing of video semantics.
- Based mostly on U-net to extract key frames of videos, implemented a learnable model to process videos, cutting them into frames and extract key frames, encoding them for transformation from sender through simulated Gaussian channel, decoding them at the receiver and reconstructing the whole video. The complete system contains one sender, one receiver and one connection node. Collaborated with Kinetics dataset to train the model and evaluate its performance.
- Achieved a compression ratio of 99.87%, video key frame restoration accuracy is over 95% under no-noise circumstance.

DEVELOPMENT AND INTEGRATION OF A-C RL-BASED ABR ALGORITHM FOR VIDEO STREAMING

Shanghai, CN

Advisor: Prof. Roger Zimmermann, School Of Computing, NUS

May 2022 - Jul 2022

- Finished an ABR algorithm based on A-C RL, designed a set of neural networks leveraging ReLU as an activation function and softmax network as an coaching network with teammates, finally received an error rate lower than 0.02.
- Led Java language to encapsulate the ffmpeg-related functions and wrapped it into a callable package to process the test video file, converted the video fragments into desired format and stream.
- Adapted the ABR model into dash.js, deployed https server to get the model, got results similar with local model.
- Awarded the honored prize among more than 30 groups (Top 10%).

ANALYSIS OF INTERNET SECURITY AND PROTECTION FROM ATTACK

Shanghai, CN

Advisor: Prof. Liyao Xiang, John Hopcroft Computer Science Center, SJTU

Jul 2021 - Feb 2022

- Built a model to protect from attacks on network under typical attacking scenario based on Gaussian Mixture Model.
- The model aims at reducing the information loss after the dataset being assailed. Based on GMM and Pareto method, the model was trained on CelebA dataset and was able to automatically classify different tags. By poisoning part of the dataset to attack the model, implemented algorithm to evaluate the performance of model on classification accuracy and cross entropy of testing texts.
- Managed reinforcement machine learning to improve the model, improved accuracy by 5%.

VISION AND ELECTROMAGNETIC GUIDENCE SOLUTIONS FOR SMART CAR

Shanghai, CN

Advisor: Prof. Bing Wang, Department Of Automation, SEIEE, SJTU

Jul 2020 - Aug 2020

- Developed a program for the smart car to recognize printed edge lines on the ground and electromagnetic wires under the track with different illuminations by using image enhancement, edge detection, and template matching.
- Designed motion control system for the smart car using SVM to predict expected speed and increase its speed with the car confined to the track. Implemented the system on a smart car with a speed of 2.8m/s, relying on camera recognition and electromagnetic wire navigation.
- Won the Eastern China's second prize of the Freescale Cup National College Student Smart Car Competition.

LANGUAGE AND IT SKILLS

- Programming Languages: C/C++, Python(mainly)
- Tools: PyTorch, TensorFlow, Keil, LaTeX, MATLAB
- Machine Learning Methods: CNN, RNN, Transformer, Diffusion Model, Large Language Model